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ORIGINAL RESEARCH

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Comparing deep learning and classical regression approaches for predicting healthcare expenditure and spending: a systematic review

John Tayu Lee^a , Melody Hsiao-San Yeh^a , Vincent Cheng-Sheng Li^a , Hsiao-Hui Chen^a , Yi-Hsuan Liu^b , Yu-Chun Chen^a and David Bin-Chia Wu^c 

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ABSTRACT

Aims: This study compares deep learning architectures with traditional regression and tree-based models for individual-level healthcare cost prediction, with particular attention to performance differences across data contexts.

Methods: We conducted a preregistered systematic review (PROSPERO CRD420251129440). Web of Science, PubMed, Embase, and Scopus were searched through August 2025. Eligible studies used real-world individual-level data (claims, electronic health records, or registries) to predict cost-related outcomes with at least one deep learning architecture and one classical regression comparator, and reported quantitative performance. Data were extracted on population, predictors, outcome horizon, model type, validation strategy, performance metrics, calibration, and interpretability.

Results: Eight studies met inclusion criteria, spanning the United States, Europe, and Asia. In longitudinal designs—such as multi-year claims prediction and medication or hospitalization time-series forecasting—sequential deep learning models, particularly LSTM and CNN-LSTM hybrids, consistently outperformed regression and tree-based algorithms. Reported gains included approximately 10–20% reductions in RMSE/MAE, R^2 improvements of 0.01–0.15, and AUROC values up to 0.78 for high-risk classification. Across studies, prior costs and utilization were consistently the strongest predictors, while social determinants and free-text features were rarely incorporated. In contrast, for low-dimensional, structured, cross-sectional medical data, generalized linear models and tree-based approaches remain robust baseline models due to their interpretability and calibration stability.

Limitations: Evidence is based on a small and heterogeneous set of eight studies, with limited external or temporal validation, short prediction horizons, and sparse assessment of calibration, economic interpretability, and fairness, warranting cautious interpretation.

Conclusions: Deep learning offers clear gains for longitudinal, sequence-rich cost forecasting, whereas tree-based methods remain highly competitive for cross-sectional tabular prediction. Overall, these findings are consistent with the proposed Complexity–Performance Hypothesis, which posits that the predictive advantages of deep learning emerge primarily when model capacity is well matched to data complexity.

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Introduction

Healthcare expenditures continue to rise globally, imposing escalating burdens on payers, governments, and patients alike^{1–3}. Accurate forecasts of individual and population-level health costs are crucial—not

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only for academic modeling but also for practical applications in insurance underwriting, risk adjustment, budget planning, and value-based payment strategies⁴. Traditional statistical approaches (e.g. linear models, generalized linear models, two-part models) frequently struggle to capture complex, nonlinear interactions in health data and may underperform when faced with high-dimensional inputs^{5–7}.

In recent years, deep learning and neural network architectures have garnered considerable attention for their promise in modeling high-dimensional, heterogeneous data^{8,9}. For example, Drewe-Boss et al. (2022) applied a deep neural network to claims data of German insurees and demonstrated superior performance over ridge regression in predicting next-year total healthcare costs¹⁰. Such applications underscore the potential of neural networks to learn latent feature interactions, temporal patterns, and nonlinearities that conventional models often miss^{7,8}.

However, despite such advances, important methodological questions remain. It is still unclear how extant studies differ in their policy or operational context, population and data sources, neural network architectures and training strategies, sensitivity analyses, software platforms, and modes of result presentation and interpretation^{11–13}. Moreover, many papers provide limited comparisons between neural network models and benchmark regression or machine learning alternatives, making it hard to judge when neural networks truly add value¹¹.

Healthcare expenditures are influenced by multiple interrelated factors. Demographic composition, particularly population aging, contributes to a growing demand for healthcare services^{14,15}. Epidemiological, lifestyle and environmental and socioeconomic factors also influence individual and population health status, are also associated with healthcare spending¹⁴. Individual healthcare spending is also shaped by utilization patterns, access to providers, and broader forces such as economic growth and technological progress^{14,15}. These determinants interact in nonlinear ways, producing highly skewed and dynamic spending distributions. Capturing such complexity requires analytical frameworks capable of modeling heterogeneity and temporal persistence in costs—motivating the application of neural network approaches to health expenditure prediction.

This systematic review aims to map the evolving landscape of neural network-based healthcare cost and expenditure prediction. Specifically, we systematically searched the literature to address the following key questions: (1) What research objectives have been addressed (e.g. total cost, cost change, high-cost classification)? (2) Which neural network methods—feedforward networks, convolutional nets, recurrent architectures (LSTM/GRU), hybrid models, or others—are used in health expenditure prediction? (3) What evaluation frameworks and result presentation strategies (e.g. error metrics, calibration, classification thresholds) are commonly employed? (4) How do neural network model performances compare with classical regression or machine learning baselines across domains and settings?

Methods

This systematic review was conducted according to a pre-specified protocol and registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number CRD420251129440).

A comprehensive search strategy was designed to identify relevant literature across multiple bibliographic databases, including Web of Science, PubMed, EMBASE, and Scopus. The search covered all publications up to August 2025. Reference lists of included studies were additionally screened to capture potentially eligible articles not retrieved through the database search.

Studies were considered eligible if they: (1) Applied deep learning or neural network algorithms (e.g. feed-forward neural network, convolutional neural network, recurrent neural network, long short-term memory [LSTM], DNN) for the prediction, estimation, or modeling of healthcare expenditure, cost, or resource use; (2) Used real-world, individual-level health data such as insurance claims, electronic health records (EHRs), administrative datasets, or registry data; (3) Reported model performance metrics (e.g. RMSE, MAE, R^2 , AUC) or compared neural network models against classical or machine-learning baselines; and (4) Were published in English.

To ensure methodological comparability, only studies that directly implemented both a deep learning (or neural network) model and at least one classical regression model (e.g. linear regression, GLM, ridge, lasso) and quantitatively compared their predictive performance were included. Additionally, studies were required to include an outcome variable explicitly representing healthcare expenditure, spending,

cost, or a closely related economic measure (e.g. medical cost, insurance charge, resource utilization expressed in monetary units).

Studies were excluded if they: (i) focused exclusively on intervention cost-effectiveness or simulation modeling without predictive application; used hypothetical, synthetic, or simulated data sources (e.g. Kaggle or other artificially generated datasets) rather than real-world data; or (iii) applied machine-learning algorithms other than neural networks without neural network comparison. (iv) did not conduct a direct comparative analysis between a classical regression model and a deep learning approach on cost- or expenditure-related outcomes.

Three conceptual domains guided the search formulation: (1) Deep learning and neural network terminology (e.g. "deep learning," "neural network*," "CNN," "RNN," "LSTM," "DNN," "artificial intelligence," "machine learning"); (2) Predictive modeling expressions (e.g. "predict*," "forecast*," "model*," "estimate*," "projection*"); and (3) Healthcare expenditure concepts (e.g. "healthcare cost*," "medical cost*," "health expenditure*," "health spending," "treatment cost*," "resource use," "resource utilization").

Alternative databases such as ACM Digital Library were evaluated but subsequently excluded due to limited coverage of health-economic literature.

During the initial screening stage, titles and abstracts were independently assessed by one trained reviewer according to pre-specified eligibility criteria¹⁶. All records that were potentially eligible or for which eligibility was unclear were retained for full-text review to avoid premature exclusion. Final inclusion and exclusion decisions were jointly discussed by at least two team members; in cases of disagreement, a third team member was consulted to resolve discrepancies and minimize the risk of systematic selection bias.

Among excluded studies, the most common reason for exclusion was that the prediction target was not healthcare cost or expenditure but alternative outcomes, such as disease incidence or hospitalisation risk. Additional studies were excluded because they did not include a direct comparison between traditional regression or tree-based models and deep learning approaches. Studies relying on simulated or synthetic data, or other non-real-world data sources, were also excluded. Based on these criteria, a total of eight studies were ultimately included, all of which used real-world data, explicitly focused on healthcare cost prediction, and directly compared deep learning models with traditional regression or tree-based approaches.

Data extraction was performed using a structured Excel framework. Extracted variables included bibliographic details, data source and setting, population characteristics, neural network architecture, input variables, model evaluation metrics, comparator models, cost or outcome definitions, and key findings.

Findings were synthesized narratively and summarized in tabular form, categorizing studies by model type, prediction task, dataset scale, and performance benchmark. Quantitative aggregation was limited to descriptive reporting due to methodological heterogeneity among studies. The systematic review followed the PRISMA 2020 reporting guidelines, and the completed checklist is provided in [Supplementary Table S1](#).

Results

The database search identified a total of 23,572 records: PubMed ($n = 1,525$), Embase ($n = 9,532$), Scopus ($n = 43$), and Web of Science ($n = 12,472$). After removing 2,164 duplicate records, 21,408 unique records were screened by title and abstract. Additionally, 11 records were identified through citation searching, all of which were retrieved and assessed for eligibility, with 10 records excluded. Of these, 21,352 were excluded as not relevant to the review objectives. 8 studies met the final inclusion criteria and were included in this systematic review. The full study identification, screening, and inclusion process is illustrated in [Figure 1](#) (PRISMA flow diagram).

Characteristics of identified studies

[Table 1](#) showed the characteristics of the identified studies. A total of eight studies met the inclusion criteria, conducted across the United States, predominantly private insurance system ($n = 5$)^{17,18,20,22,23}, Europe, including Germany, which has single-payer systems ($n = 1$)¹⁰, and France, which has universal

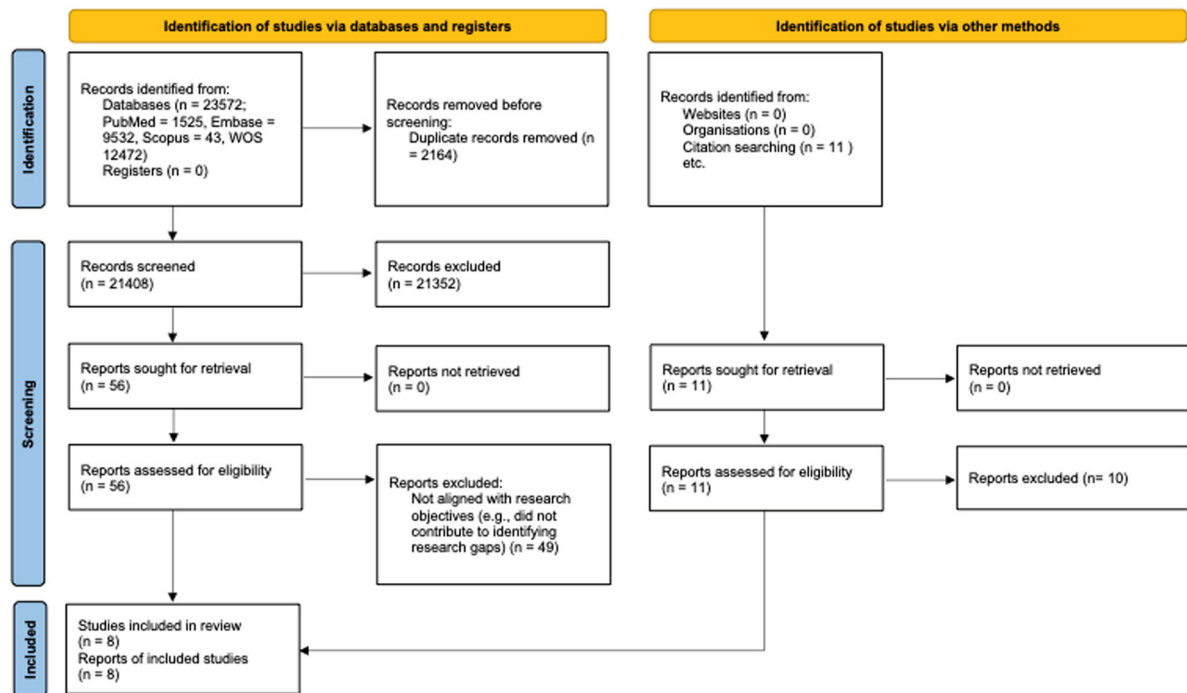


Figure 1. PRISMA flow diagram of study identification.

health insurance with multiple payers ($n=1$)²¹; and Asia (China = 1)¹⁹. Sample sizes ranged from approximately 50,000 to over 1.4 million individuals, encompassing data sources such as commercial insurance claims ($n=3$)^{17,18,23}, national health insurance databases ($n=3$)^{10,20,21}, and electronic medical records (EMR) ($n=1$)¹⁹, while one study used retail-level pharmacy transaction records ($n=1$)²².

Six studies adopted longitudinal designs (multi-year or time-series forecasting)^{10,17,18,20,22,23}, while two were cross-sectional cost-regression studies^{19,21}. Table 2 also shows the outcome variables primarily included: (1) Total annual or per-member-per-month expenditure ($n=5$)^{10,17,18,20,21}, (2) Medication-specific or pharmacy expenditures ($n=3$)^{18,22,23}, (3) Preventable cost¹⁷, and (4) In-hospital costs ($n=1$)¹⁹.

In the heart failure population study by Lewis et al. (2021), preventable costs were defined on the basis of the follow-up period and operationalized using insurance claims data, in which hospitalizations and emergency department visits classified as potentially avoidable—according to clinical quality indicators and emergency care categorization algorithms—were aggregated into total reimbursed costs¹⁷. Among the eight included studies, only this study explicitly focused on preventable costs as the primary economic outcome, whereas the remaining studies primarily examined total medical expenditures or in-hospital costs as the main economic indicators¹⁷.

Deep learning architectures varied across studies: feed-forward networks (DNN/MLP; $n=2$)^{10,21}, sequence-based models (LSTM, stacked LSTM, CNN-LSTM, RNN; $n=4$)^{17,18,20,23}, hybrid or multi-view networks ($n=1$)¹⁹, and variance-based generative adversarial network (V-GAN; $n=1$)²². Comparators included classical regression approaches—such as GLM, ridge, and LASSO regression ($n=6$)^{10,17,18,20,21,23}—and tree-based machine-learning methods including Random Forest (RF), Gradient Boosting Machine (GBM), or CatBoost ($n=4$)^{17,20–22}. Model performance was most commonly evaluated using R^2 or adjusted R^2 ($n=5$)^{10,18,20,21,23}, MAE ($n=3$)^{18,19,21}, RMSE ($n=5$)^{18–20,22,23}, and AUC ($n=2$)^{10,17}; several studies additionally reported precision@k, cost-capture, or correlation coefficients for discrimination and calibration assessment ($n=3$)^{10,17,19}.

Clinical and policy contexts of prediction

Based on the comparative models in Table 2, this section shows the methodological frameworks and model performance of each included study. One U.S. study focused on population health management, predicting preventable hospitalizations, emergency visits, and expenditures among heart failure patients

Table 1. Characteristics of included studies on deep learning applications for healthcare expenditure prediction.

Study	Journal	Country	Study population and sample size	Aim and objectives	Study design	Temporal design
Lewis et al. (2021) ¹⁷	Scientific Reports	United States	93,260 adults with heart failure continuously enrolled for two years (2016–2017) in a large U.S. commercial insurance database	To compare deep learning approaches (sequential and non-sequential) with traditional logistic regression for predicting preventable hospitalizations, emergency department (ED) visits, and healthcare spending among heart failure (HF) patients To develop and evaluate a deep neural network for predicting one-year future health-care costs from longitudinal claims data, and to compare its performance with ridge regression and other benchmark models for both total cost prediction and cost-change identification.	Retrospective prognostic modeling using 12 years (2006–2017) of commercial claims data; patients divided into training, validation, and testing sets (7:2:1).	Longitudinal (12 years)
Drewe-Boss et al. (2022) ¹⁰	BMC Medical Informatics and Decision Making	Germany	1,403,346 insured individuals from the InGef database (> 60 German statutory health insurances, 2010–2017), representative of the German population by age, sex, and state of residence.		Retrospective population-based model-development and comparison study using longitudinal claims records (2010–2015 observation period; 2016–2017 evaluation period).	Longitudinal (multi-year)
Kaushik et al. (2020) ¹⁸	Applications of Machine Learning – Chapter 14:	United States	Over 50,000 patients with prescription-based pain medication claims between January 2011 and April 2015 from the Truven MarketScan Commercial Claims and Encounters database (U.S.)	To forecast weekly and monthly patient medication expenditures using advanced deep-learning time-series methods, and to compare multi-headed CNN–LSTM ensemble architectures against traditional forecasting models.	Retrospective, observational, longitudinal cost-forecasting study using five years of patient-level claims data; 70/20/10 train-validation-test split.	Longitudinal (2011–2015)
Yu et al. (2021) ¹⁹	Knowledge-Based Systems	China	750,000 discharged patient records from Chinese hospitals, drawn from a large-scale electronic medical records (EMR) database including diagnosis texts, operation codes, and patient demographics.	To predict in-hospital resource utilization, specifically estimating in-hospital costs and length of stay (LOS) using electronic medical records via a deep learning approach.	Retrospective observational study using structured and unstructured EMR data for supervised learning. The study introduced the RUP (Resource Utilization Prediction) framework integrating multiple data views (e.g. structured features, diagnosis text, operation text) into a unified model.	Cross-sectional (within-stay)
Yang et al. (2018) ²⁰	BioMed Eng OnLine	United States	Medicaid-enrolled patients in Texas, including both general population and high-cost high-need (HCHN) subgroups, analyzed using several years of longitudinal claims data. Sample size not explicitly reported, but the dataset contained millions of monthly member records covering multiple years of expenditure history.	To evaluate the predictive performance of machine learning models (including recurrent neural networks) for forecasting future healthcare expenditures, with a focus on identifying and managing high-cost high-need (HCHN) Medicaid patients. The study aimed to quantify temporal persistence in healthcare spending and to benchmark deep learning models against classical regression approaches.	Retrospective longitudinal cohort study using administrative claims data. Each patient's historical expenditures, diagnoses, procedures, and medications were used to predict next-period (12-month) expenditures.	Longitudinal (multi-year)

(continued)

Table 1. Continued.

Study	Journal	Country	Study population and sample size	Aim and objectives	Study design	Temporal design
Vimont et al. (2022) ²¹	European Journal of Health Economics	France	General population covered by the French National Health Data Information System (SNDIS), using a 1/97 representative random sample of the national database for the year 2016.	To compare the predictive performance of machine learning (Random Forest and Neural Network) models with a classical Generalized Linear Model (GLM) in forecasting individual healthcare costs using nationwide claims data. The study sought to determine whether machine learning offers meaningful gains over traditional regression in health cost modeling.	Retrospective, cross-sectional, population-based study using routinely collected administrative claims data from SNDIS. Separate models were trained for the entire population and for high-cost subgroups.	Cross-sectional (single year 2016)
Kaushik et al. (2020) ²²	IEEE Access	United States	Patients purchasing a specific pain medication; the exact sample size not specified but derived from retail-level transaction records (time-series data).	To develop and test a variance-based generative adversarial network (V-GAN) model that minimizes the difference in variance between model and actual data for predicting medicine expenditures	Retrospective computational modeling study using prescription expenditure time-series data.	Longitudinal, time-series forecasting model using sequential expenditure data over multiple time points.
Kaushik et al. (2017) ²³	Proceedings of the IEEE International Conference on Machine Learning and Data Science (MLDS)	United States	>50,000 patients in the United States purchasing a prescription-based pain medication, extracted from the Truven MarketScan insurance claims dataset (2011–2015)	To forecast monthly per-patient expenditure on a prescription-based pain medication using deep learning (LSTM) and to compare performance against a traditional statistical model (ARIMA).	Retrospective observational study using secondary insurance claims data.	Longitudinal time-series design—first 48 months (2011–2014) used for training, final 12 months (2015) used for testing.



Table 2. Comparative findings of deep learning and classical models for healthcare expenditure prediction.

Study	Outcome variable (e.g. target variable)	Deep learning model type	Other model compared	Features included in the model	Model performance metrics	Findings	Identified challenges
Lewis et al. (2021) ¹⁷	Preventable hospitalizations (1) Preventable hospitalizations (6-month horizon); (2) Preventable ED visits (6-month horizon); (3) Preventable costs (1-year horizon)	Sequential Deep Learning: Long Short-Term Memory (LSTM) with attention mechanism; Convolutional Neural Network (CNN). Non-sequential Deep Learning: Feed-Forward Neural Network (FNN).	Traditional logistic regression (two variants with limited features); Enhanced logistic regression (939 features); Gradient Boosting Model (GBM).	939 knowledge-driven features (demographics, comorbidities, procedures, medications, costs, utilization trends) and data-driven features generated from claims using Word2Vec embeddings (11-year histories, both sequential and non-sequential).	Sequential Deep Learning achieved highest AUROC: 0.778 (hospitalizations), 0.681 (ED visits), 0.727 (costs). Precision at top 1%: 39–43% (vs. 12–15% for logistic regression).	Deep learning models (particularly LSTM and CNN) outperformed all logistic regression baselines and GBM in discriminating patients at high risk for preventable events and spending. Prior utilization (hospitalizations, LOS, ED visits) and comorbidities (COPD, CKD, anxiety) were key drivers. Demonstrated that deep learning can enhance risk stratification for targeted interventions in value-based HF care.	– Lack of socioeconomic, laboratory, and EHR variables limited interpretability and completeness. – Preventable outcomes based on existing algorithms (Billings, PO) may misclassify events. – No external validation beyond random internal split. – Potential inefficiency due to 11-year data window. – Limited transparency and explainability of deep learning models.
Drewe-Boss et al. (2022) ¹⁰	Total one-year health-care costs (Euros) and cost change categories (increase or decrease >100-fold between periods).	Fully-connected multilayer perceptron (4 hidden layers; 50 neurons each, ReLU activation, dropout 0.25, ADAM optimizer, ensemble of 5 runs).	Ridge regression	High-dimensional representation of claims per quarter (24 quarters): ICD-10-GM diagnoses, ATC drug codes, DRG, OPS procedures, GOP billing codes, physician specialty keys (F-G), sex, age, and sector-specific costs → 333,024-dimensional feature vector per patient after frequency filtering.	Pearson correlation (r), Spearman (ρ), Mean Absolute Prediction Error (MAPE, ϵ), R^2 , Cumming's Prediction Measure (CPM), Area under Precision-Recall curve (AUPRC) and AUROC for cost-change classification.	The deep network substantially outperformed ridge regression and traditional baselines for overall cost prediction and cost-change identification. Performance improved with larger training data and longer observation windows. Integrated gradients revealed that deep learning relied more on clinically meaningful diagnosis (ICD-10) and drug (ATC) codes than billing codes, suggesting enhanced medical interpretability.	– Model interpretability and uncertainty estimation remain limited despite using integrated gradients. – No external validation outside the InGet dataset. – Model may be affected by changes in billing regulations or coding practices. – No explicit inclusion of social determinants or environmental factors. – Further research needed on optimal architectures and generalization to other phenotypes.
Kaushik et al. (2020) ¹⁸	Average patient-level healthcare (or medication) expenditures per time period (weekly/monthly).	Multi-Headed LSTM (MHLSTM)/Multi-Headed CNN (MHCNN)/Hybrid CNN-LSTM ensemble. Stacked LSTM model.	Auto Regressive Integrated Moving Average (ARIMA)	Demographics (age, gender, region), number of prescriptions, comorbidity indicators, drug utilization rates, and previous expenditure trends.	Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 (Coefficient of Determination).	Multi-headed ensemble architectures significantly improved accuracy and robustness in forecasting patient-level expenditures. The hybrid CNN-LSTM ensemble effectively captured both short-term temporal and long-term trend dependencies, showing potential for healthcare cost forecasting and resource planning.	– Computationally intensive due to multiple heads and ensemble structure. – Model interpretability limited; lacks clear feature attribution. – Validation restricted to medication-related expenditures from Truven dataset (no general health-system costs). – No inclusion of social or behavioral determinants. – Suggested integrating explainable AI (e.g. SHAP) and testing on broader expenditure categories in future work.

(continued)

Table 2. Continued.

Study	Outcome variable (e.g. target variable)	Deep learning model type	Other model compared	Features included in the model	Metrics used for evaluating model performance	Model performance metrics	Findings	Identified challenges
Yu et al. (2021) ¹⁹	Two continuous targets: in-hospital costs (CNY) and length of stay (days). Both were jointly predicted via multi-task learning.	RUP model (multi-view, multi-task neural network) – combines a record encoder (using embedding + attention mechanisms) and a resource predictor to capture heterogeneous data sources. Each view (structured, diagnosis text, operation text) was encoded separately and integrated through attention-based fusion.	Linear Regression (LR), Support Vector Regression (SVR), Decision Tree (DT), M5 Model Tree, Random Forest (RF), Gradient Boosting Machine (GBM)	Structured data: Age, sex, hospital department, admission type, operation indicators, discharge status, and admission diagnosis code. Unstructured data: Diagnosis and operation texts from EMRs encoded via word embeddings. Derived features: Aggregated operation counts, department-level indicators, and resource group codes.	Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Pearson Correlation Coefficient (PCC).	The RUP deep learning model achieved: – RMSE: 0.693 (cost), significantly lower than all baselines ($p < 0.001$). – MAE: 0.451. – PCC: > 0.85 for cost and > 0.80 for LOS. Compared to GBM and RF, the RUP model reduced prediction error by ~12–15%.	The multi-view deep learning approach (RUP) outperformed traditional regression and tree-based models across all metrics, demonstrating its capability to integrate textual and structured data effectively. The inclusion of diagnosis/operation texts and attention-based fusion was critical for performance gains.	– Limited generalizability beyond Chinese EMR systems. – Computational cost of multi-view and multi-task deep learning training. – Limited explainability: RUP provides limited insight into how textual features affect predictions. – Absence of external validation across hospitals or countries. – Potential bias due to missing data or coding inconsistencies across EMR systems.
Yang et al. (2018) ²⁰	Per Member Per Month (PMPM) healthcare expenditure—continuous variable representing total future cost per patient.	Recurrent Neural Network (RNN) capturing temporal dependency and sequence information across multiple months of prior cost and utilization data.	Linear Regression (LR), LASSO Regression, and Gradient Boosting Machine (GBM) – all benchmarked against RNN for identical data splits.	Demographic characteristics (age, gender, region), clinical features (diagnosis codes, procedure codes, medication use), historical cost and utilization patterns, and comorbidity indicators. Prior costs over multiple periods (lags of 1–4 months) were included to evaluate temporal persistence.	R-squared (R^2), Root Mean Squared Error (RMSE), and comparison of prediction error distributions across models.	RNN outperformed GBM and LASSO in prediction accuracy. R-squared values greater than 0.7; lower prediction error for HCHN patients compared to the general population. RNN performance improved when incorporating 3–4 prior cost periods, showing its ability to capture time-dependent cost patterns.	The study found strong temporal persistence in individual healthcare expenditures, particularly among high-cost, high-need patients. The RNN model outperformed classical regression (LASSO, Linear Regression) and ensemble models (GBM), highlighting the advantages of sequential learning for dynamic healthcare spending prediction. It also emphasized that temporal features (e.g. recent spending trajectories) were more predictive than static patient demographics or diagnoses.	– Lack of transparency and interpretability in RNN model outputs (black-box limitation). – Absence of socioeconomic variables beyond claims-based features. – Limited generalizability outside the Medicaid population. – No fairness or equity analysis included. – Computationally intensive training and limited discussion of hyperparameter tuning.
Vimont et al. (2022) ²¹	Annual individual healthcare expenditure in 2016 (continuous variable, in euros).	Feed-forward Neural Network with nonlinear activation functions, trained on patient-level covariates.	Generalized Linear Model (GLM) (gamma distribution with log link) and Random Forest (RF).	Age, gender, pre-existing conditions, Charlson comorbidity index, healthcare utilization (consultations, hospital stays, medications, imaging), and prior-year total healthcare costs.	Adjusted R-squared ($\text{adj-}R^2$), Mean Absolute Error (MAE), and Hit Ratio (HIR)—the percentage of correctly identified high-cost individuals in the top decile of actual expenditure.	RF achieved an $\text{adj-}R^2$ of 0.475, MAE of €1,338, and HIR of 67%. GLM had an $\text{adj-}R^2$ of 0.347, MAE of €1,635, and HIR of 58%. Neural network had an $\text{adj-}R^2$ of 0.316, MAE of €1,660, and HIR of 55%.	– Random Forest: $\text{adj-}R^2 = 0.475$, MAE = €1,338, HIR = 67%; – GLM: $\text{adj-}R^2 = 0.347$, MAE = €1,635, HIR = 58%; – Neural Network: $\text{adj-}R^2 = 0.316$, MAE = €1,660, HIR = 55%.	– Neural network underperformance attributed to small sample and limited optimization. – Lack of temporal features (cross-sectional data only). – Possible overfitting in RF due to large variable set. – Model explainability and transparency remain limited for ML approaches. – Results may not generalize to other healthcare systems or cost structures.

(continued)

Table 2. Continued.

Study	Outcome variable (e.g. target variable)	Deep learning model type	Other model compared	Features included in the model	Metrics used for evaluating model performance	Model performance metrics	Findings	Identified challenges
Kaushik et al. (2020) ²²	Daily average expenditure on a specific pain medication.	Variance-Based Generative Adversarial Network (V-GAN) integrating an LSTM-based generator and a CNN-based discriminator. Multilayer Perceptron (MLP), and standard Long Short-Term Memory (LSTM) network	Linear Regression (LR), Gradient Boosting Regression (GBR)	Patient demographics and historical purchase/expenditure time-series data; no mention of clinical or socioeconomic covariates	Root Mean Square Error (RMSE)	The V-GAN achieved the lowest RMSE compared to all benchmark models, demonstrating superior predictive accuracy. (Exact RMSE not numerically reported in text but indicated as the minimal among comparators.)	The V-GAN framework successfully reduced variance discrepancies between real and generated expenditure distributions, outperforming traditional regression and deep learning baselines (LSTM, MLP). The model proved particularly effective for modeling non-stationary healthcare expenditure series.	– Absence of external validation and unclear representativeness of the medication dataset – limited to a single therapeutic area and cost type – lack of socioeconomic and clinical features restricts generalizability to broader healthcare spending – model interpretability not discussed.
Kaushik et al. (2017) ²³	Average monthly patient expenditure on a specific pain medication (out-of-pocket spending, excluding insurer-paid amount).	(1) Standard (vanilla) LSTM with one hidden layer (7 memory cells) and (2) Stacked LSTM with two LSTM layers (4 memory cells each).	ARIMA (Auto Regressive Integrated Moving Average, benchmark)	Univariate time-series data of monthly medication expenditures; demographic variables (age, gender, region, birth year) were available but not used in main models (proposed for future multivariate work). Demographics available (age, gender, region, birth year), but not used in main models (future work proposed multivariate models)	Root Mean Square Error (RMSE) R ² (Coefficient of Determination)	ARIMA: RMSE = 16.855, R ² = 0.8045 Standard LSTM (7 memory cells): RMSE = 14.617, R ² = 0.8048 Stacked LSTM (4 + 4 memory cells): RMSE = 13.693, R ² = 0.8159	Both LSTM models outperformed ARIMA. Stacked LSTM achieved the best predictive performance. LSTMs were more robust, nonparametric, and better at capturing temporal dependencies. Implication: LSTMs could support forecasting tasks in healthcare expenditure and policymaking.	– Model limited to univariate forecasting (no demographic or socioeconomic covariates). – Lack of external validation across medication types or populations. – Results based on a single drug category (pain medication), limiting generalizability. – Future work proposed developing multivariate LSTM models incorporating demographics and expanding to multiple medication categories for broader healthcare cost forecasting.

using LSTM and CNN models to support risk stratification and care coordination under value-based care frameworks¹⁷. Two studies from Germany and France examined risk adjustment and insurance premium prediction, using large national claims datasets to forecast annual healthcare costs and assess whether deep or tree-based models improved fairness and predictive accuracy in reimbursement and capitation formulas^{10,22}. Both were conducted within statutory health insurance systems, emphasizing model transparency and calibration for regulatory use^{10,21}. A U.S. Medicaid study analyzed high-need, high-cost (HCHN) populations, applying recurrent neural networks to predict per-member-per-month expenditures²⁰. A Chinese study using electronic medical records focused on hospital resource utilization, predicting in-hospital costs and length of stay with a multi-view DNN that combined structured and unstructured clinical data¹⁹. This work supports hospital efficiency improvement and DRG-based payment reform, linking predictive analytics to provider-level financial management¹⁹. Finally, three U.S. studies addressed pharmaceutical expenditure forecasting, using claims or retail transaction data to predict weekly, monthly, or daily medication spending^{18,22,23}.

Neural network architectures in the included studies

Neural network architectures were applied according to input type and temporal structure. Feedforward deep neural networks handled static or high-dimensional features, capturing nonlinear relationships in cross-sectional data. Recurrent networks (RNNs/LSTMs) modeled temporal dependencies in longitudinal or multi-wave sequences. More complex architectures—including hybrid CNN-LSTM models, multi-head ensembles, and generative frameworks (V-GANs)—were used for heterogeneous or non-stationary data, integrating structured and unstructured inputs.

Across the studies, neural network design generally reflected data characteristics (see Table 3). Feedforward DNN models were applied to static features²¹, while Vimont et al., 2022²¹ included a simple neural network alongside statistical models for comparison. Recurrent architectures captured temporal dependencies when explicitly implemented, with convolutional neural networks (CNNs) and fully connected networks (FCNs) used to process spatial and static features, respectively (e.g. Lewis et al. 2021¹⁷). Hybrid or generative models addressed complex temporal or heterogeneous inputs (Kaushik et al. 2017²³; 2020²²; Yu et al. 2021¹⁹). However, some studies (e.g. Yang et al. 2018²⁰) primarily relied on traditional machine learning approaches.

Evaluation frameworks and performance metrics

All eight studies evaluated model performance using quantitative metrics aligned with their clinical or policy objectives^{10,17–23}. Six studies employed internal multi-split validation (train–validation–test or cross-validation)^{17–19,21–23}, while two—Drewe-Boss et al. (2022) and Yang et al. (2018)—used temporal holdout samples to assess generalizability^{10,20}. None performed full external validation across independent datasets.

Longitudinal prediction studies ($n=6$)—Drewe-Boss et al. (2022), two studies from Kaushik et al. (2020), Kaushik et al. (2017), Yang et al. (2018), and Lewis et al. (2021)—focused on forecasting accuracy and discrimination^{10,17,18,20,22,23}. Most reported RMSE ($n=4$)^{18–20,22,23} and some reported MAE ($n=2$)^{10,18} and R^2 or correlation coefficients (r , ρ) to quantify fit. Lewis et al. (2021) also included AUROC, precision@k, and cost-capture to evaluate high-risk identification¹⁷, while Drewe-Boss et al.

Table 3. Neural network architectures applied in the included studies.

Neural network architecture	Data type	Models	Studies
Feedforward/DNN	Static or high-dimensional data	Standard deep neural network	Drewe-Boss et al. 2022 ¹⁰
Simple neural network	Static or high-dimensional data	Simple neural network (comparison with statistical models)	Vimont et al., 2022 ²¹
Recurrent	Longitudinal or time-dependent data	RNN, LSTM (auxiliary: CNN, FCN)	Lewis et al. 2021 ¹⁷
Hybrid/generative	Complex temporal patterns, heterogeneous data	Hybrid or generative models (e.g. V-GAN)	Kaushik et al. 2017 ²³ ; 2020 ¹⁸ ; 2020 ²² ; Yu et al. 2021 ¹⁹

(2022) and Kaushik et al. (2020) added AUPRC and Mean Absolute Percentage Error (MAPE), respectively, for trend and cost-change detection^{10,18}. Sequential models such as LSTM and CNN-LSTM were assessed primarily on temporal forecasting stability and error reduction^{18,23}.

Cross-sectional studies ($n=2$)—Vimont et al. (2022) and Yu et al. (2021)—emphasized model fit and numerical error for static cost prediction^{19,21}. Vimont et al. (2022) used adjusted R^2 , MAE, and Hit Ratio to compare neural networks with GLM and Random Forest for national reimbursement modeling²¹. Yu et al. (2021) evaluated RMSE, MAE, and Pearson correlation coefficient for in-hospital cost and length-of-stay prediction¹⁹.

Across all papers, R^2 /adjusted R^2 ($n=5$), MAE ($n=4$), RMSE ($n=5$), and AUC or related discrimination metrics ($n=2$) were most frequently reported^{10,17–23}. Only one study addressed calibration to a limited extent¹⁹, and none included decision-curve, net-benefit, or fairness analyses.

Explainability frameworks such as Integrated Gradients was carried out in Drewe-Boss et al. (2022)¹⁰.

Comparative performance of neural networks versus classical models

Across studies comparing neural architectures with classical regression and ensemble methods, neural models generally performed best when applied to longitudinal or temporally rich healthcare data^{10,17–23}. Recurrent and hybrid architectures captured temporal persistence and nonlinear dynamics that traditional approaches often missed. In a U.S. Medicaid study, Yang et al. (2018) showed that, for predicting raw Per Member Per Month (PMPM) dollar amount expenditures, a recurrent neural network increased test R^2 from about 0.14–0.17 (for linear regression and GBM) to 0.30 and reduced RMSE by roughly one-third, with similar improvements in RMSE among the top 10% highest-cost patients; on log- and percentile-transformed outcomes, R^2 differences across models were smaller, but the RNN still achieved lower overall and high-cost error.²⁰ Kaushik et al. (2017) likewise found that stacked LSTMs improved RMSE by nearly 19% over ARIMA forecasts for a medication-spending time series²³, while a later multi-headed CNN-LSTM, Convolutional LSTM, and CNN-LSTM ensemble achieved even stronger performance with R^2 values around 0.8–0.9 in patient-level forecasting¹⁸. Lewis et al. (2021) also reported that sequential deep models outperformed gradient boosting and logistic regression in predicting preventable hospitalizations and costs, attaining AUROCs up to approximately 0.78.¹⁷

The picture was more mixed in cross-sectional cost prediction. In a French nationwide claims study, Vimont et al. (2022) found that a relatively shallow two-hidden-layer MLP performed similarly to a generalized linear model but was clearly outperformed by a random forest (adj- R^2 0.475 vs. 0.347 and 0.316 for GLM and neural network; MAE €1,338 vs. €1,635 and €1,660), suggesting limited benefit of shallow neural architectures on tabular cost data²¹. By contrast, in China, Yu et al. (2021) demonstrated that when rich structured and textual EMR inputs were available, a multi-view deep learning model integrating diagnosis and operation text with structured features clearly surpassed all regression and tree-based baselines, reducing prediction error by roughly 12–15%.¹⁹

Figure 2 illustrates roughly how predictive performance evolves as data become richer and more complex. The x-axis moves from simple cross-sectional claims to increasingly heterogeneous, longitudinal, and multimodal data; the y-axis represents relative model performance. Classical regression performs well for simple, low-dimensional tabular inputs but tends to plateau as complexity increases. Tree-based machine-learning models (e.g. random forests, gradient boosting) extend performance by capturing nonlinearities, typically outperforming regression across most settings. It is important to note that Figure 2 is a conceptual framework developed to synthesize the evidence reviewed in this study and to illustrate the proposed Complexity–Performance Hypothesis²⁴. The figure is intended to visualize the overall performance trends of different model families as data complexity increases and does not represent empirical estimates derived directly from any individual study.

Shallow feed-forward neural networks (green) show the weakest performance in the cross-sectional regime—consistent with Vimont et al. (2022), where a two-layer MLP achieved an adjusted R^2 of only 0.316, below both RF (0.475) and GLM (0.347)²¹. Without sequential structure or rich feature representations, shallow networks offer limited advantage over traditional methods. However, deep neural architectures (red) behave differently. Even in cross-sectional settings, when richer input modalities are available—as in the multi-view RUP model using structured and textual EMR features (Yu et al. 2021)—

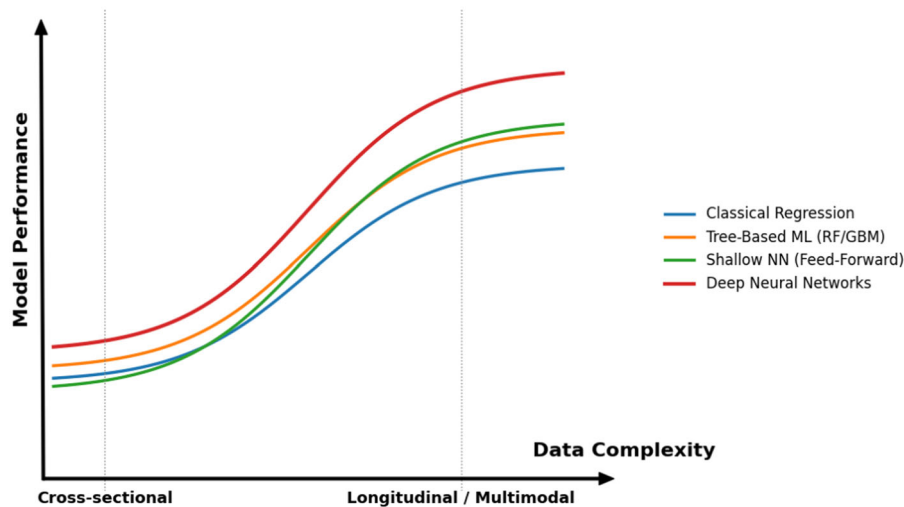


Figure 2. Model performance by data complexity.

deep learning can clearly outperform both regression and tree-based baselines.¹⁹ As data become increasingly high-dimensional, sequential, or multimodal, deep networks benefit from their higher capacity and representation learning capabilities. This produces the strongest gains in the longitudinal/multimodal regime, where studies such as Lewis et al. (2021), Yang et al. (2018), Kaushik et al. (2017/2020), and Kaushik et al. (2020 V-GAN) show that recurrent, convolutional, and hybrid deep models outperform classical and ensemble methods by substantial margins^{17,18,20,22,23}.

Building on these findings, we formulate what we term the Complexity–Performance Hypothesis, which posits that model accuracy is maximized when the capacity of the modeling approach is matched to the complexity of the data. When data are simple and homogeneous, classical regression performs competitively, reflecting its efficiency in low-dimensional settings. As data gain nonlinear structure or moderate dimensionality, tree-based machine-learning methods begin to offer clear advantages by capturing interactions and flexible decision boundaries. With further increases in data richness—such as large-scale, temporal, or multimodal inputs—DNN models provide the greatest performance gains, often surpassing alternative approaches by a wide margin.

Discussion

Primary findings

Across the eight included studies, deep learning models demonstrated consistent performance gains over classical regression, though the magnitude and nature of improvement varied by study design and data type. The strongest advantages were observed in longitudinal analyses, while cross-sectional applications yielded smaller or mixed results. These results provide empirical support for the Complexity–Performance Hypothesis, which proposes that predictive accuracy depends on how well a model’s architecture is matched to the underlying structure and complexity of the data.

In longitudinal or time-series prediction tasks, such as forecasting healthcare expenditures, hospitalizations, or medication costs, neural networks with sequential structures—particularly LSTM and CNN-LSTM—achieved notable accuracy improvements, typically showing roughly 10–20% lower prediction error (RMSE or MAE) and absolute R^2 improvements of about 1–15 percentage points compared with classical regression or autoregressive models. These architectures effectively captured temporal persistence, nonlinear cost dynamics, and patterns of high-cost episodes that static models failed to represent.

In contrast, cross-sectional models predicting single-year or aggregated costs showed more limited improvements. Shallow feed-forward neural networks performed similarly to or worse than classical regressions—while tree-based methods such as random forests and gradient boosting often matched or exceeded neural performance in overall fit and calibration. However, when unstructured or multimodal inputs such as EMR text were incorporated, deep learning achieved clearer advantages, yielding larger

error reductions and demonstrating greater flexibility in modeling heterogeneous clinical information. Taken together, these findings suggest that deep learning delivers its greatest advantages in settings with temporal depth or multimodal complexity, whereas classical and ensemble methods can remain competitive—or superior—when the task is cross-sectional and inputs are purely tabular or the neural architecture is relatively shallow.

Across studies, feature importance patterns were highly consistent. Prior costs and utilization emerged as the dominant predictors, followed by comorbidities, procedures, and medication use. Very few models incorporated social determinants, behavioral data, or text-based clinical variables, limiting interpretability and potential policy translation.

When compared across methodological families, deep learning outperformed classical regression in nearly all contexts, but tree-based models remained highly competitive in cross-sectional settings that emphasize transparency and calibration. Overall, the evidence suggests that deep learning excels when data are longitudinal, high-dimensional, or sequential, while classical and tree-based methods continue to provide stable, interpretable baselines for policy-oriented modeling such as risk adjustment and premium setting.

Although most included studies involved large sample sizes, often ranging from tens of thousands to several million observations, these studies generally possessed sufficient statistical power to detect meaningful performance improvements, such as significant reductions in RMSE or MAE. Nevertheless, the overall conclusions remain influenced by contextual factors, including the concentration of studies within specific countries or insurance systems and the reliance on single-database observational designs. Importantly, these limitations do not overturn the overarching conclusion that deep learning offers advantages over traditional regression in longitudinal time-series cost prediction. However, effect size estimates—particularly across specific populations, subgroups, or neural network architectures—should be interpreted with caution.

Through the integration of theoretical considerations and empirical evidence, this review articulates the Complexity–Performance Hypothesis as a unifying framework to explain these findings. In data scenarios characterized by relatively low dimensionality or limited complexity, traditional regression models are often sufficient to achieve effective prediction. As data complexity increases—through greater nonlinearity or interaction effects—tree-based machine learning models begin to demonstrate advantages in flexibility and fit. When data are both high-dimensional and longitudinal or multimodal, DNN models with hierarchical feature-learning capabilities exhibit the greatest potential to improve overall prediction error and identify high-cost tails. Importantly, this hypothesis is derived inductively from the eight included empirical studies and is presented as a conceptual framework to guide model selection across data structures and application scenarios, rather than as a precise estimate of a single causal effect.

Implications for health economics and policy

From a health economics and healthcare finance perspective, these findings translate into operational guidance regarding when deep learning is most appropriate for real-world application. Specifically, deep learning models are most relevant for healthcare cost prediction tasks that integrate longitudinal data or multiple data modalities²⁵. Under such conditions, neural networks tend to achieve lower overall prediction error and demonstrate superior ability to identify high-cost populations, compared with traditional regression and some tree-based models. In practice, this improved discrimination can support earlier identification of high-risk individuals and inform budget planning and resource allocation decisions²⁶.

When sufficient data infrastructure, computational resources, and institutional support are available, deploying high-complexity deep learning models can be cost-effective and analytically justified. By contrast, in cross-sectional risk-adjustment or insurance estimation settings—where available predictors are limited to standard claims data or a small number of structured covariates—the empirical evidence indicates that shallow neural architectures yield only marginal improvements in explanatory power or calibration. In such scenarios, tree-based machine learning models often achieve comparable or superior performance, while offering greater interpretability.

Moreover, under regulatory and governance constraints, tree-based models typically impose lower communication and transparency costs when engaging with policymakers and supervisory agencies²⁷. Accordingly, GLM-based two-part models and selected tree-based approaches remain appropriate baseline choices for many applied health economics analyses. Adoption of deep learning in these contexts should be

contingent upon direct head-to-head comparisons conducted on identical datasets and data partitions, demonstrating reproducible and policy-relevant improvements in cost capture, budget accuracy, and high-cost population monitoring.

Finally, practical deployment of deep learning models requires careful consideration of downstream resource requirements, including computational infrastructure, MLOps maturity, and the capacity to explain model behavior to decision-makers and regulators²⁸. Within health economics, the Complexity–Performance Hypothesis therefore implies not only a statistical matching between model capacity and data structure, but also a balanced assessment of predictive gains, transparency, and institutional feasibility. Given that most routine analytic workflows in insurance and public health agencies continue to rely on GLM and tree-based models, the limited availability of readily deployable deep learning platforms remains a non-trivial barrier to widespread adoption²⁹.

Future directions and fundamental challenges

To guide future research, this review delineates five fundamental challenges in applying deep learning to healthcare expenditure prediction, outlines actionable strategies for their resolution, and identifies priorities for collaboration between machine learning scientists and health economists (see Table 4).

Challenge 1: limited economic interpretability

Despite the high predictive accuracy of deep learning architectures, their opaque internal representations constrain economic interpretation and policy translation³⁴. The “black-box” nature of neural networks obscures causal pathways and cost-driver interactions, limiting their relevance for accountability and decision-making in health financing²⁷. To align model outputs with economic reasoning, deep learning models should integrate explainable-AI (XAI) approaches—such as SHAP, Integrated Gradients, or Layer-wise Relevance Propagation—and report marginal effects comparable to those derived from generalized linear or two-part models. Developing hybrid “interpretable-deep” frameworks that embed econometric constraints within neural architectures may reconcile predictive precision with transparency, facilitating their integration into budget-impact and resource-allocation analyses³⁵.

Table 4. Challenges of deep learning in predicting healthcare spending and expenditure, and how they may be overcome with future development.

Challenges	Current status	Strategies
Limited economic interpretability	Deep learning models achieve high predictive accuracy but provide limited transparency for economic reasoning, policy translation, or causal inference.	• Incorporate explainable AI (XAI) frameworks such as SHAP or Integrated Gradients. • Report marginal effects and sensitivity analyses comparable to traditional regression outputs. • Develop hybrid interpretable-deep models tailored for cost drivers and budget impact interpretation.
Insufficient benchmarking against classical regression	Many studies lack direct, standardized comparisons between deep learning models and established regression frameworks (GLM, two-part, or hurdle models) ³⁰ .	• Require side-by-side benchmarking using identical datasets and performance metrics. • Emphasize calibration, fairness, and welfare implications beyond accuracy. • Promote transparent reporting of incremental predictive and economic gains.
Short temporal prediction horizons ³¹	Most models predict one-year costs, overlooking long-term expenditure dynamics or persistence of high-cost status.	• Extend modeling horizons to multi-year predictions. • Incorporate longitudinal claims data and time-dependent survival or sequence models. • Evaluate temporal generalization <i>via</i> rolling-window validation.
Limited data diversity and generalizability	Current studies rely heavily on single-country or proprietary datasets, restricting generalization across systems and populations ³² .	• Promote multi-country, multi-payer benchmarking datasets. • Integrate open, de-identified EHR or administrative data with harmonized coding. • Implement domain adaptation and federated learning for privacy-preserving collaboration.
Underuse of social and behavioral determinants	Models predominantly use demographic and clinical variables, neglecting social determinants of health (SDOH) and behavioral predictors ³³ .	• Link cost data with area deprivation, socioeconomic, or behavioral risk indices. • Quantify incremental explanatory power of SDOH variables on expenditure prediction. • Integrate SDOH embeddings in multimodal architectures.

Challenge 2: insufficient benchmarking against classical regression

The absence of standardized benchmarking between deep learning models and traditional regression frameworks remains a major methodological gap³⁶. Many studies neglect formal comparisons with generalized linear, hurdle, or two-part models, or apply inconsistent data partitions and performance metrics, precluding rigorous assessment of incremental value. Future research should direct, head-to-head benchmarking using identical datasets, covariates, and validation procedures³⁶. Evaluation must extend beyond statistical accuracy to encompass calibration, discrimination, and welfare-relevant metrics such as cost-capture or decision-curve analysis³⁷. Transparent reporting of incremental explanatory power relative to GLM baselines will clarify when the additional complexity of deep learning models is justified³⁸.

All eight studies included in this review met the pre-specified minimum comparison criteria by incorporating at least one traditional or tree-based comparator^{10,17–23}. However, only Drewe-Boss et al. (2022) and Vimont et al. (2022) partially satisfied more stringent benchmarking requirements, including consistent sample splits direct comparison with GLM two-part models^{10,21}. The remaining studies reported results under heterogeneous subgroup definitions or outcome specifications, limiting their suitability as complete benchmarking cases^{17–20,22,23}.

Drawing on principles articulated in DECIDE-AI, we emphasize the importance of clearly defining application scenarios, target populations, comparator models, and evaluation structures a priori³⁶. Although not specific to economic evaluation, these principles provide a useful framework for structuring systematic and transparent benchmarking between neural networks and classical cost-prediction models.

Challenge 3: short temporal prediction horizons

Most existing models focus on one-year prediction horizons, reflecting budget-cycle pragmatism but neglecting the persistence and temporal dependencies of high-cost conditions³¹. Extending prediction windows to multi-year horizons using longitudinal claims and EHR data would better capture dynamic health-state transitions³⁹. Sequence-based architectures (e.g. LSTM, GRU, transformer models), combined with rolling-window validation, should be used to assess temporal stability³⁹.

Across the included studies, training windows often exceeded the prediction horizon. For example, Drewe-Boss et al. (2022) used data from 2010–2015 to predict costs in 2016–2017¹⁰, while Lewis et al. (2021) analyzed commercial claims data spanning 2006–2017.¹⁷ Kaushik et al. (2017, 2020a, 2020b) typically employed four to five years of training data with the final year reserved for testing^{18,22,23}, whereas Vimont et al. (2022) and Yu et al. (2021) relied primarily on cross-sectional designs^{19,21}. However, details regarding training and observation periods were often confined to tables and not fully discussed in the main text. Future studies should provide explicit disclosure of calendar years and observation windows to strengthen assessment of longitudinal performance and cross-context generalizability.

Challenge 4: limited data diversity and generalizability

Most studies rely on single-country, single-payer, or proprietary datasets, limiting external validity and constraining cross-system generalization⁴⁰. Differences in coding standards, benefit structures, and reimbursement schemes further impede model transferability. Addressing this challenge requires the establishment of multi-country, multi-payer benchmark datasets with harmonized variable definitions and cost metrics⁴¹. Privacy-preserving and federated learning architectures can facilitate cross-institutional collaboration without compromising confidentiality⁴¹. Domain-adaptation techniques should be developed to recalibrate models for heterogeneous populations and health systems. Publicly accessible, de-identified datasets adhering to FAIR (Findable, Accessible, Interoperable, Reusable) data principles are essential for reproducibility and methodological transparency⁴².

Challenge 5: underuse of social and behavioral determinants

Current deep learning cost-prediction models overwhelmingly depend on demographic and clinical variables, with limited incorporation of social determinants of health (SDOH) or behavioral risk factors. Yet socioeconomic conditions, education, and lifestyle behaviors exert substantial influence on utilization patterns and cost trajectories⁴³. Integrating SDOH requires linking administrative and clinical data with census, deprivation,

or behavioral datasets, and representing these inputs through multimodal embeddings that capture latent social context⁴⁴. Quantifying the incremental explanatory and policy value of SDOH variables will help identify structural cost drivers and guide equitable allocation of healthcare resources.

Study limitations

This systematic review included only eight studies, and substantial heterogeneity was observed in research design, data sources, healthcare systems, outcome definitions, and performance metrics. As a result, quantitative meta-analysis was not feasible, and findings were synthesized narratively by identifying consistent directional patterns. Most studies relied on single-country, single-database designs and lacked external validation across time periods or healthcare systems. Additionally, the predominance of one-year prediction horizons limits assessment of long-term robustness and generalizability. Consequently, the Complexity–Performance Hypothesis should be viewed as a working hypothesis, requiring further systematic validation rather than a definitively established causal mechanism.

Conclusion

Although the studies reviewed offer preliminary support for the Complexity–Performance Hypothesis, rigorous empirical validation is still in its early stages. Future research should conduct head-to-head benchmarking of modeling approaches on the same datasets, systematically varying dimensionality and temporal depth to determine the threshold at which adopting deep neural networks becomes justified over classical regression or ensemble methods. Deep learning offers clear gains for longitudinal, sequence-rich cost forecasting, whereas tree-based methods remain highly competitive for cross-sectional tabular prediction. Overall, these findings are consistent with the proposed Complexity–Performance Hypothesis, which posits that the predictive advantages of deep learning emerge primarily when model capacity is well matched to data complexity.

Transparency

Declaration of funding

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Author contributions

J.T.L. conceived and designed the study, conducted the analysis, and led the manuscript writing. M.H.-S.Y. performed the literature search and screening. All authors contributed to drafting and refining the of the manuscript. All authors read and approved the final manuscript.

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Use of generative AI

The authors used ChatGPT (OpenAI, GPT-4o, 2025) to assist with language polishing and grammar/word checking only.

Data availability statement

This systematic review is based solely on data extracted from published articles. No new datasets were created. Extracted data tables and the PRISMA checklist can be provided upon reasonable request.

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